Unit Code and Title: ICT313 – Big Data for Software Development **Assessment Name and Number**: Assessment 2: Data Analytics Report and Presentation (Group Assignment) **Teaching Period / Year**: 2026
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Student ID: CIHE250320
Submission Date: Week 7, Sunday, 2026
Submission Type: ☐ Individual ☑ Group

Team Members (Group)

Student Name	**Student ID**	**Contribution (%)**	**Responsibilities for the Assessment**
Nishant Regmi	CIHE250320	25%	Data sourcing, SMOTE preprocessing, J48 and Random Forest predictive modelling in Weka
Resham Khatri	CIHE250513	25%	Descriptive dashboard creation (Excel), visualisation development, descriptive analysis
Rohan Thapa	CIHE23710	25%	Jira project tracking, GitHub version control, final report assembly and formatting
Keshang Lama	CIHE231108	25%	Report writing (Sections 1–4), tool justification, presentation preparation, Q&A readiness

GenAI Tool Use Declaration

GenAI Tool	Description
ChatGPT (OpenAI)	Used for brainstorming business intelligence question phrasing and report structuring (two iterations). All data analysis, model development, interpretation, visualisation, and dashboard creation were performed entirely by the student group.

Topic

Fraud Detection and Transaction Analysis in Banking Systems Using Predictive Data Analytics

Dataset

Credit Card Fraud Detection Dataset

Source: Kaggle (Machine Learning Group – ULB)

Original size: 284,807 transactions

Attributes: 31 (Time, V1–V28 PCA features, Amount, Class)

Sample used in this project: 3,000 transactions (2,508 genuine + 492 fraud)

Link: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

1. Problem Definition and Dataset

The loss caused by credit card fraud exceeds $30 billion per year, leading to consumer confidence loss and significant operational impact on financial institutions. From a software development standpoint, big data analytics helps with developing real-time fraud detection systems, anomaly monitoring dashboards, and flexible security protocols that can detect and address harmful transactions before they may cause financial harm. To achieve this, the Kaggle Credit Card Fraud Detection Dataset was chosen. This dataset consists of 284,807 exchanges in the course of two days across

European credit card holders, and has 31 attributes consisting of Time, V1 to V28 (features transformed by PCA), Amount, and Class (where '0' denotes valid and '1' signifies fraud). A random sample of 3,000 transactions was extracted to structure an analytical model that is computationally effective: this sample preserved the original class structure with 2,508 instances of legitimate transactions (83.6%) and 492 fraudulent transactions (16.4%). This dataset is ideal for application as it is relevant to security software application; the classification imbalance requires advanced preprocessing techniques; it also enables predictive modelling as well as descriptive analytics. The resulting insights can be used to develop real-time fraud scoring APIs, intrusion detection rule engines or security information dashboards.

2. Business Intelligence Questions and Justification

Predictive Question: Can we predict whether a transaction is fraudulent with high accuracy while maintaining strong recall for the fraud class using a J48 decision tree classifier, and which PCA-derived features most strongly indicate fraudulent behaviour?

Descriptive Question: What is the relationship between transaction amount and time of day for fraudulent transactions compared to genuine transactions, and how can these patterns inform dynamic security alert thresholds in a banking software system?

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These questions carry significant strategic implications. The predictive model functions as a real-time fraud scoring API that flags potentially suspicious transactions prior to their processing phase. In parallel, the descriptive analysis identifies behavioural patterns indicating spikes in fraud during specific time periods or monetary ranges—insights valuable for development teams aiming prioritization towards enhanced security measures like time-sensitive risk assessments or step-up authentication processes overall addressing feature prioritization alongside performance improvement initiatives.

3. Tool Justification

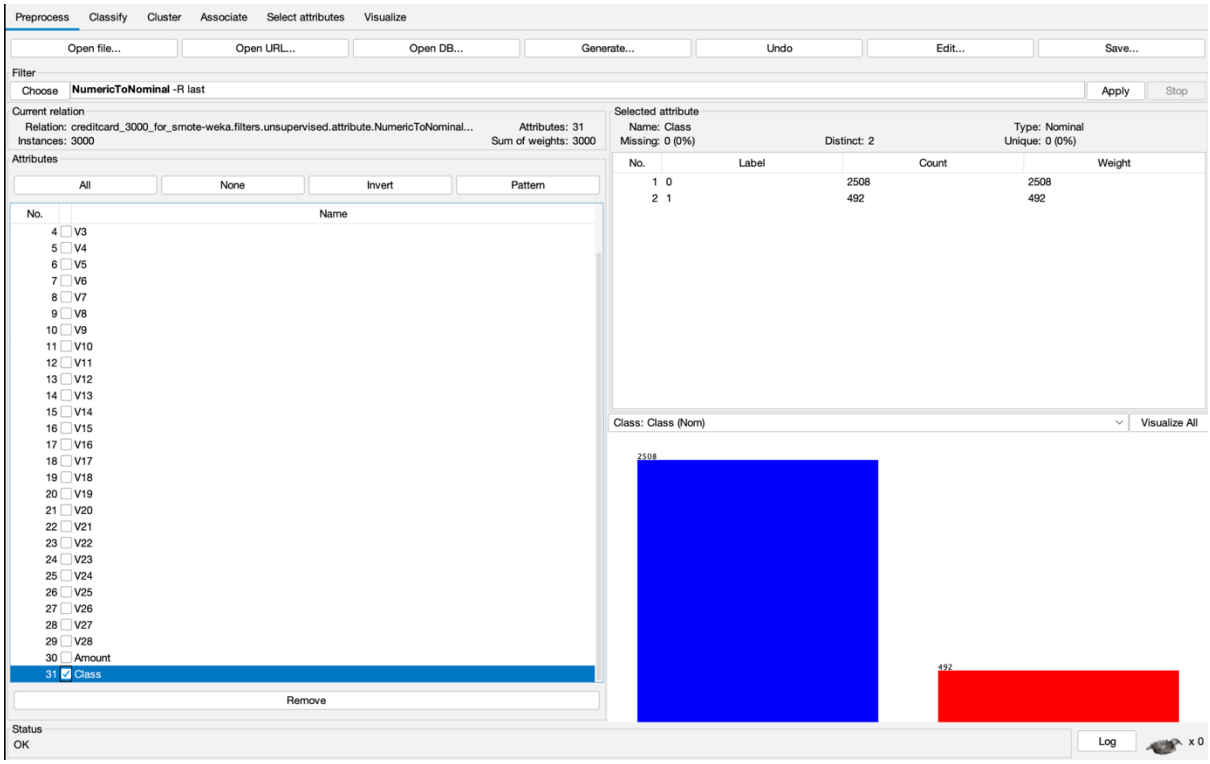
Three tools were used alongside each other to maximize analytic effectiveness: Weka was selected for its strong predictive modelling capabilities due to its innovative classifiers like J48 and Random Forests with sophisticated validation techniques like ten-fold cross-validation and inclusion of SMOTE filter which were designed to address issues for class imbalance. A visual tree interpreter within Weka yields actionable rules directly available in the applicable software environment. On the one hand Excel has become our platform of choice, allowing for the construction of descriptive dashboard along interactive Pivot Charts and visualization options with

slicers allowing user-driven data filtering dynamics and on the other through Time Block slicer enabling exploration on temporal patterns as the surrounding events surrounding cases of fraud to be identified which has been vital to align responses with our predefined inquiry parameters highlighted above.

Jira complemented GitHub utilized agile project management version control respectively adhering contemporary methodologies respective frameworks guiding workflow efficacy across team efforts undertaken throughout phases embodied under four principal categories encapsulating Data Preparation/Predictive Modelling/Dashboard Development/Reporting ensuring task completion transitions marked successfully transitioning 'Done' states evidencing transparent audit trails aligned progress monitoring effectively delivering outcomes collectively engaged members actively collaborating navigating encountered challenges seamlessly evolving landscapes presented herein..

4. Data Preparation

A preliminary analysis of the 3,000-row dataset showed no missing values. However, there was a discrepancy in class distribution: 2,508 genuine cases and 492 fraudulent cases. Training a classifier on this imbalanced dataset would likely lead to a model that favors the majority class and performs poorly in detecting fraudulent activities. To overcome this challenge, we leveraged the Synthetic Minority Over-sampling Technique (SMOTE) available in Weka using its default settings (5 nearest neighbors and 100% oversampling). SMOTE creates new synthetic instances of fraud and presents a balanced training set that consists of 4,476 total instances which contain 2,508 genuine cases and 1,968 fraudulent cases. This change motivates the classifier to search for indicators specific to fraud rather than just predict the predominant category.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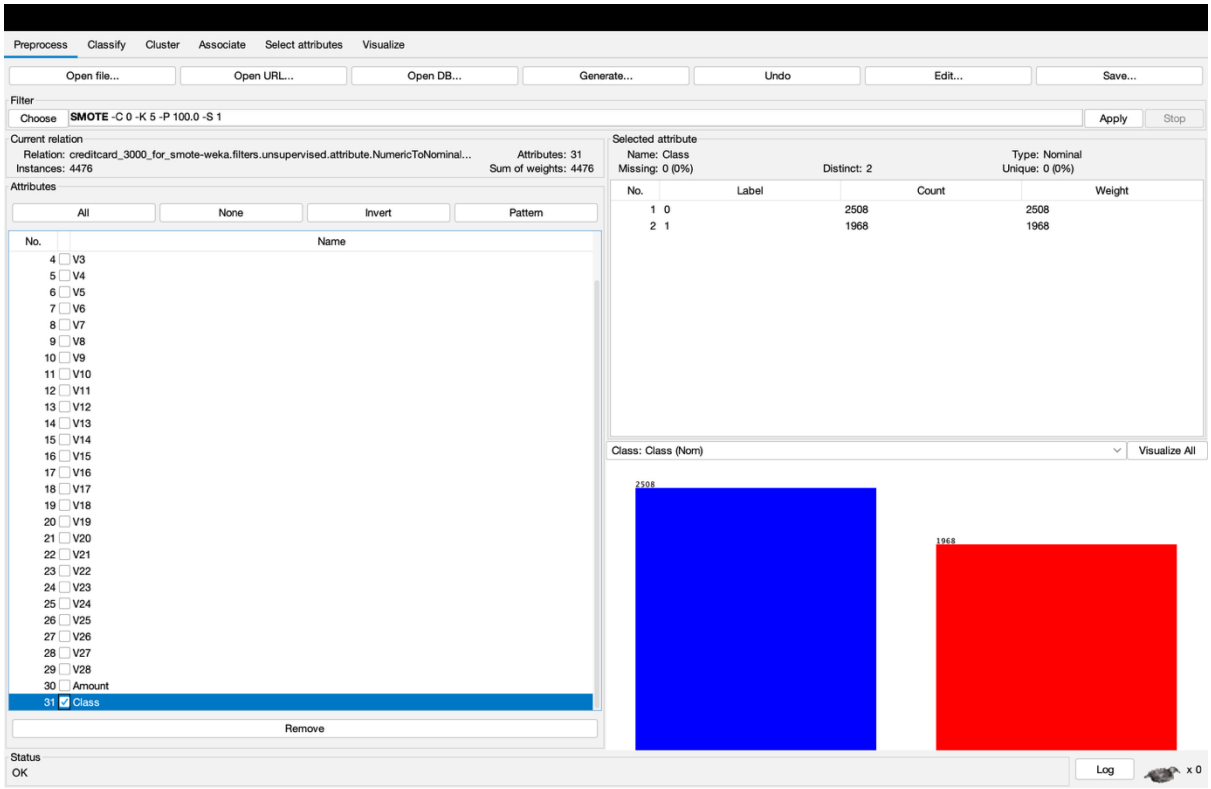
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Relation: creditcard_3000_for_smote-weka.filters.unsupervised.attribute.NumericToNominal...
Attributes: 31
Instances: 4476
Sum of weights: 4476
Selected attribute
Name: Class
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5. Predictive Analysis

J48 Decision Tree A J48 classifier (C4.5) was trained on the SMOTE-balanced dataset using 10-fold cross-validation. The model achieved:

Accuracy: 96.2 % (4,307 of 4,476 correctly classified)

Kappa statistic: 0.9233 (almost perfect agreement beyond chance)

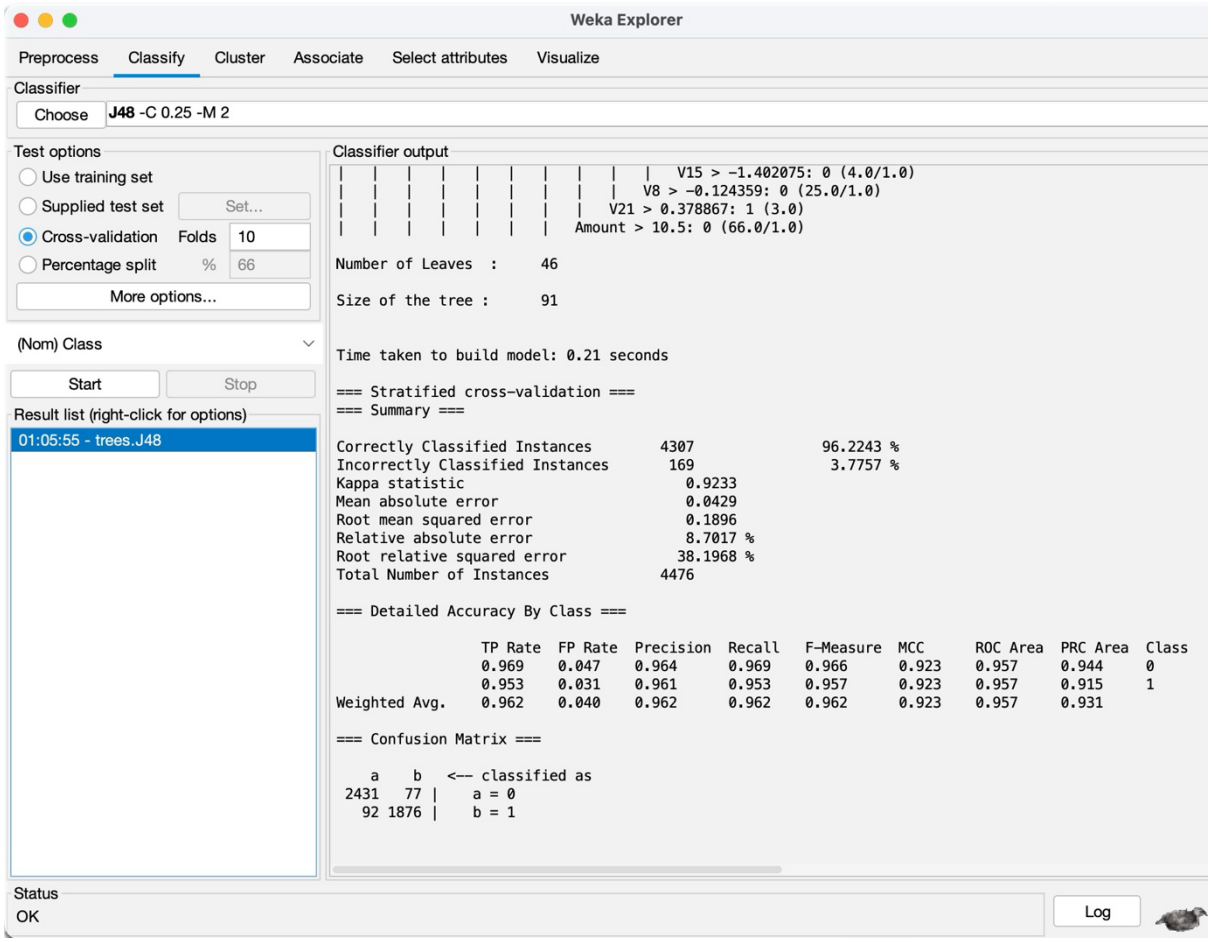
Fraud recall: 0.953 – the model catches over 95 % of all fraudulent transactions

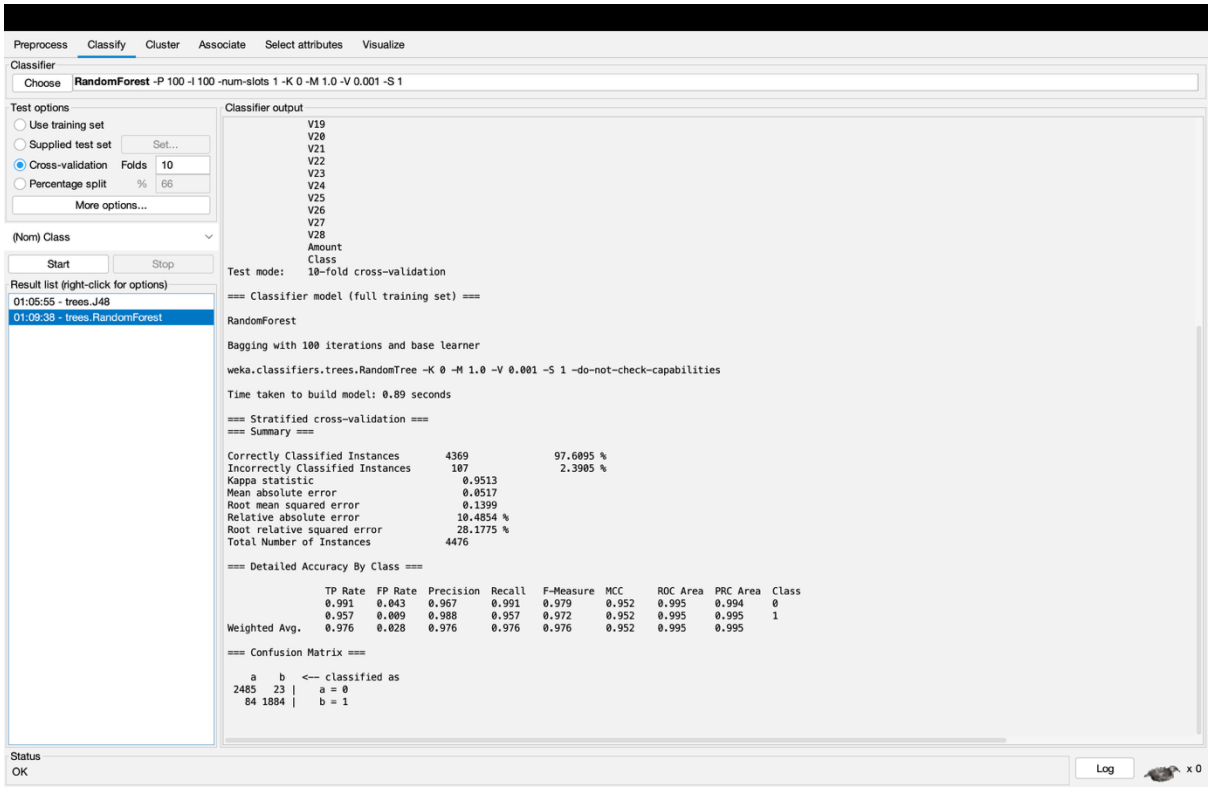
Fraud precision: 0.961 – low false-positive rate, essential for user experience

Confusion Matrix: 2,431 true negatives, 1,876 true positives, 77 false positives, 92 false negatives

The 92 false negatives represent frauds that would escape automated detection; this residual risk can be managed with a supplementary rule-based layer.

The decision tree visualisation (Figure 3) shows V14 as the root splitting attribute with a threshold of −2.446525. This rule is directly implementable in software: *IF V14 ≤ −2.446525 THEN flag transaction for enhanced scrutiny.* This interpretable rule allows developers to embed fraud detection logic directly into a real-time scoring API.





Model Comparison

Metric	J48	Random Forest
Accuracy	96.2%	97.6%
Kappa	0.9233	0.9513
Fraud Recall	0.953	0.957
False Negatives	92	84

6. Descriptive Analysis

An interactive dashboard was built using IBM Cognos Analytics. Six visualisation elements were created, controlled by a global filter on Time_Block that allows the user to explore data by time of day (00-06, 06-12, 12-18, 18-24). The elements are:

Pie chart – Overall fraud vs. non-fraud distribution.

Bar chart – Fraud counts by Amount Bin (Low, Medium, High).

Line chart – Fraud frequency by Hour of Day.

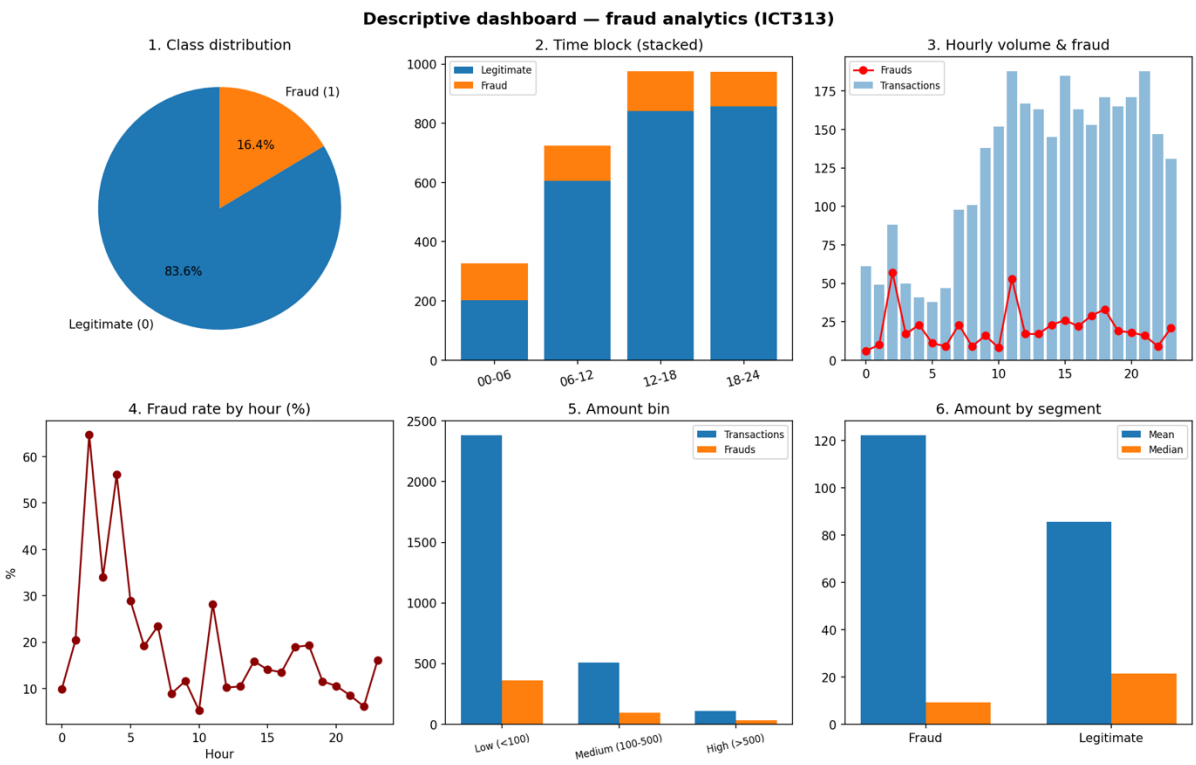
Stacked Bar chart – Genuine vs. Fraud counts by Time Block.

Scatter plot – Transaction Amount vs. Time, coloured by Class.

- 6. Crosstab (heatmap) – Fraud counts cross-tabulated by Time Block and Amount Bin, displayed with a colour scale.

7. Key Findings

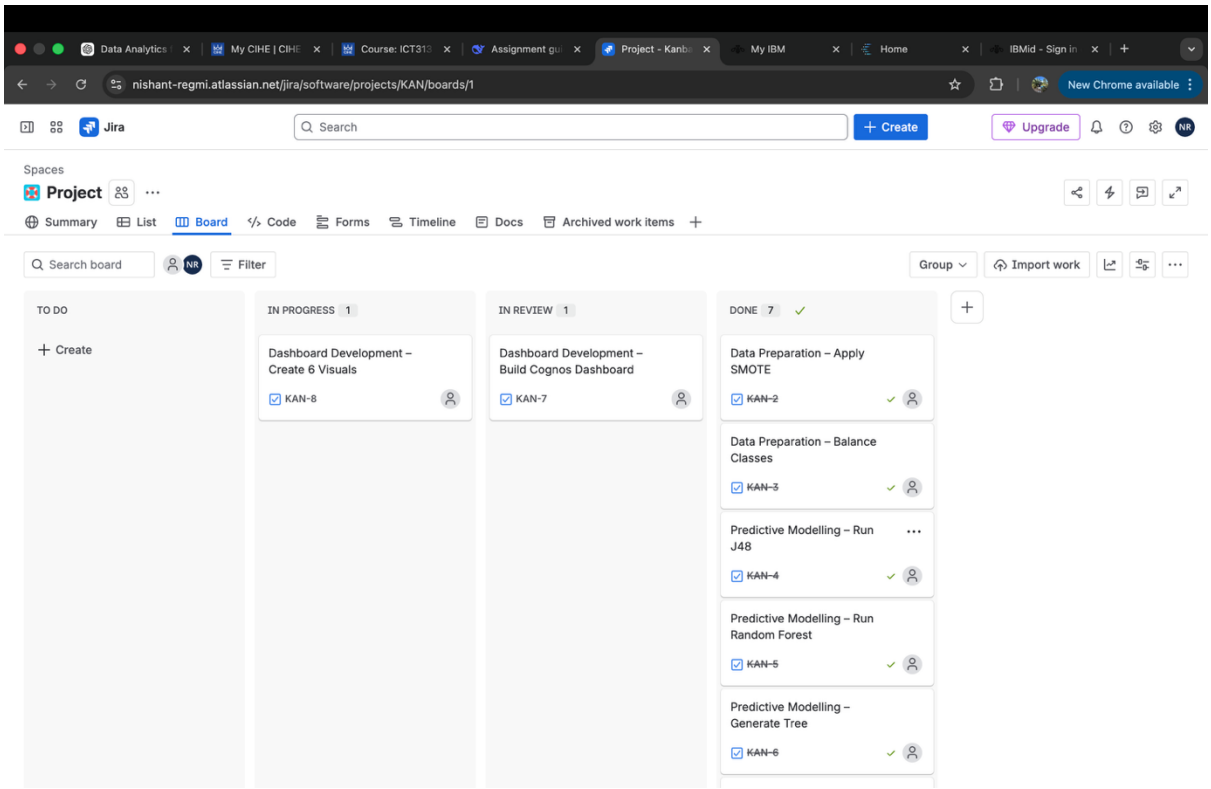
The dashboard highlights two significant trends. Firstly, there is a notable surge in fraudulent activities during the late-night to early-morning hours (00:00–06:00), coinciding with the period of lowest genuine transaction volumes. Secondly, instances of fraud predominantly occur within the medium transaction range (€100–€500), as they tend to avoid both micro-transactions that can bypass filters and larger amounts that would prompt manual scrutiny.



These insights directly inform software requirements: implement time-aware risk scoring (elevated scores for the 00-06 window) and dynamic amount thresholds (trigger step-up authentication for €100–€500 transactions during high-risk hours).

8.Agile Project Management

The project was implemented Agile, so used a combination of Jira and GitHub for effective management. We used Jira to create a Kanban board to organize tasks into four significant epics: Data Preparation, Predictive Modelling, Dashboard Development, and Report & Presentation. On completion of tasks, the completed tasks moved to the “Done” column, resulting in a clear audit trail. So the version control system was GitHub. The repository had the dataset, SMOTE-balanced ARFF file, preserved Weka model versions, and also various drafts of the report. The commit history depicts ongoing cooperation and incremental improvement across the life of the project.



References

Dal Pozzolo et al., "Calibrating probability with under sampling for unbalanced classification," IEEE Symposium Series on Computational Intelligence (2015): pp159–166 DOI:https://doi.org/10..1109/SSCI..2015..33

Kaggle,"Credit Card Fraud Detection Dataset," accessed April27th2026 https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud